

Riya Vazirani Laheja

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EDUCATION

RIDGE HIGH SCHOOL

Basking Ridge, NJ

Expected Graduation: June 2028; **Weighted GPA:** 4.66; **Honors:** Principal's honor roll

STEM Classes: AP Physics 1, Honors Precalculus, Honors Chemistry, Honors Biology, Honors Algebra II, Computer Science Principles

TECHNICAL SKILLS: Python, C/C++, HTML, CSS, JavaScript, GitHub, Deepnote, Arduino

Clubs: Girls Who Code (VP), Science League (Chem/Bio/Olympiad), Amnesty International (President), Mu Alpha Theta Honors Society, Comp Sci Club (Education), STEMxx (President), Business Club (Officer)

Other Activities: JV Soccer (midfield), Violin (Pit Orchestra - Cinderella The Musical, Chamber Orchestra)

OBJECTIVE

Passionate about applying analytical thinking, technical and computational skills, and creative problem-solving to develop impactful, innovative engineering and software solutions.

AREAS OF INTEREST: Computer Engineering, Artificial Intelligence, Materials Science, Robotics

ACADEMIC ENRICHMENT PROGRAMS

2025-Present	COLUMBIA UNIVERSITY SCIENCE HONORS PROGRAM (SHP) ● Selected into Columbia's competitive weekend STEM program ● Courses: <i>Bioengineering Design, Applied Data Structures in Python</i>	Manhattan, NY
Spring 2025	BWSI MIT SPRING PROGRAM ● Selected into the competitive BWSI MIT Spring Program ● Courses: Aspiring Engineers, Digital Logic and Computer Design - explored AI, machine learning, internet infrastructure, web hosting, and collaborative tools like GitHub and Deepnote, Boolean logic and arithmetic, basic memory elements	
Summer 2025	GIRLS WHO CODE PATHWAYS PROGRAM ● Learned mini concepts in: Cybersecurity, Web Development, Machine Learning, hands on learning in: Python, HTML, CSS, JS ● Immersion experience at JP Morgan	

HONORS/AWARDS

2025	CONGRESSIONAL APP CHALLENGE (COMPETITION VIDEO LINK) ● Awarded Honorable Mention for SymptoScan, a medical symptom detection app designed for use on any device to help users identify potential health concerns	Basking Ridge, NJ
2025 - 2026	BIOLEAGUE: Top 5% in New Jersey (14th/276); CHEM LEAGUE: 1st/366	Basking Ridge, NJ
2024 - 2026	VOLUNTEERING: Math/violin tutoring, tech teens at Bernards Township ● Recipient of the 250 hour gold pin award at Ridge High School x 2	Basking Ridge, NJ
2026	VOICE OF DEMOCRACY (VFW): 1st place audio essay on patriotism (ESSAY)	Bernardsville, NJ
2024-2026	HACKATHONS ● Top 8 at Ridge Hackathon (2025): Program that tracked emission during trash transport ● 1st place award in BRHS Hacks (2024): Won by programming two games ● Top 10 at Ridge Hackathon (2026): Mystery science game with mini games for clues	Somerset County, NJ

PROJECTS/ RESEARCH EXPERIENCE

Spring 2025	AI SLEEP PREDICTOR (LINK): MIT BWSI SPRING ● Built a machine learning model that predicts optimal sleep hours based on various factors ● Used Python and Jupyter-style notebooks in Deepnote ● Integrated basic random forest prediction and analysis
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June - Aug 25	MIT MANTIS LAB	
	Implemented bubble-sizing visualization controls + agentic workflow (MIT Kellis Lab)	
May- Aug 25	ARDUINO CIRCUIT PROJECTS	
	- Storming Robots: Designed, coded, assembled multiple interactive electronic projects - STEM Tec Expo at Ride High School: Built fully functioning Arduino board that could produce sound with piezo	
2025-Present	GITHUB PORTFOLIO (PERSONAL WEBSITE)	
	Developed and maintained several repositories showcasing diverse projects	
2022 - 2024	INSPIRIT AI: APPLIED AI RESEARCH PROGRAMS	
	- Trained and evaluated machine learning models using Python and Google Colab - Conducted data preprocessing, feature selection, and model performance analysis	

EXTRACURRICULAR ACTIVITIES

2025-Present	SCIENCE OLYMPIAD	Basking Ridge, NJ
	Selected to compete in regional competition in CAD and Codebusters	
2024-Present	GIRLS WHO CODE (VICE PRESIDENT)	Basking Ridge, NJ
	- Led and taught Python, AI, Web Dev workshops for middle and elementary schoolers - Helping run meetings, fundraise money, gather guest speakers, promote club in school	
2024-Present	STEMXX (PRESIDENT)	Basking Ridge, NJ
	- Helping run the meetings and manage the club's social media - Frequently organizing labs and putting together informational slideshows for the club	
2024-Present	AMNESTY INTERNATIONAL (PRESIDENT)	Basking Ridge, NJ
	Planned and organized AmnesTEA that generated over \$600 for donation	
2024-Present	THE START SMALL MOVEMENT (TSSM)	Basking Ridge, NJ
	<i>Co Founder of Startup</i>	
	Launched an initiative to help female owned small businesses - providing social media support, content creation and outreach (blogs, website planning, etc)	
2022-Present	SOCCER (MIDDLE AND HIGH SCHOOL)	Basking Ridge, NJ
	- Junior Varsity team midfield at Ridge High School for 2 years - Won girls state championships in both 7th and 8th grade	
2023-2025	STORMING ROBOTS (COMPUTER/ROBOTICS EDUCATION OUTSIDE SCHOOL)	Branchburg, NJ
	- Advanced robotics and algorithm program focused on problem-solving and deep logic - Developed several C/C++ projects emphasizing recursion, search/sort algorithms, and efficient code design	
2015-Present	TAE KWON DO (SECOND DEGREE BLACK-BELT)	Basking Ridge, NJ
	- Learning discipline, focus, self-defense, and cooperation with peers; breaking boards, forms, and endurance conditioning - Selected for leadership program as an instructor	

SERVICE EXPERIENCE

Summer 2025	GROUP COUNSELOR (JUNE 30- AUG 1, 2025)	Basking Ridge, NJ
	- Selected for summer camp group counselor position for elementary school aged children - Created close bonds with the children and was an enjoyable and fulfilling experience	
2024-Present	WHEELS (NON-PROFIT)	
	Summer intern: Published 4 blogs - different types of health awareness globally	
2024-2026	VSY (VEDIC SAMAJ YOUTH)	Basking Ridge, NJ
	Director of Web Development: Maintaining, developing, and keeping the website - Taught and led coding camp for elementary and middle school students to learn Python	